MCB Consulting Group (MKTG 411)

Fall 2026

Meeting Times

Section	Time	Days	Room	Building
001	2:00p - 3:15p	TTh	1216	TNRB
002	3:30p - 4:45p	TTh	2325	TNRB

CAO: Cameron Bale

- Contact: cameron.bale@byu.edu
- Office Hours: By appointment.

Senior Analytics Partners:

- Izzy Smith
  - Contact: is047894@student.byu.edu
  - Office Hours: M/W 3:30-4:30pm, Th 1:00-1:30pm (in-person, Blue Line)
- Paige Flake
  - Contact: paigeef@student.byu.edu
  - Office Hours: M/W 9:30-10:45am, (in-person, Blue Line)

Overview

Congratulations on joining the MCB Consulting Group! In this job, you will learn to use analytics to generate marketing insights. Employees will learn the fundamentals of R and marketing analytics and apply them in a series of projects. Skills learned on the job build on (and complement) data analysis techniques from marketing research (MKTG 401) and introductory business analytics (MSB 325); prior programming experience is helpful but not required.

Assignments include exercises, knowledge checks, and projects. Success requires consistent study and practice. Although we can cover only a selection of techniques, the materials and projects will prepare interested employees to continue developing their analytics skills.

Job Purpose

My goal is to help employees become capable analysts, reflective thinkers, and principled decision makers. Employees will combine business acumen with technical skills to identify business problems, propose data-driven solutions, and communicate results in managerially relevant terms.

Training (Learning) Outcomes

Learning on the job is a shared responsibility. I will offer my knowledge and experience; you commit to arrive spiritually, mentally, and physically ready to learn and contribute. Employees who do so can achieve the following outcomes:

- Increased faith in Jesus Christ and the confidence to conduct analyses in R and communicate practical, business-relevant results.
- The ability to analyze with integrity by documenting decisions, identifying limitations in data and models, and quantifying uncertainty.
- The motivation to improve continually and take responsibility for mistakes.
- The ability to recognize and cultivate excellence in their own analytical work and that of others.

Tools and Materials

Employees will analyze data in R, a free, open-source language widely used in analytics. Fluency in R is a marketable and transferable skill.

Bring a personal or BYU-rented laptop to every meeting for coding and assignments. Job materials will be available on Canvas; download them if you want access after your employment ends.

R

Install the following tools in order:

1. Install R, the language that runs your code.
2. Install RStudio Desktop, the integrated development environment (IDE) used to write and run R code.

Supplementary Material

Two free books provide useful support:

- *R for Data Science (2e)* by Hadley Wickham, Mine Çetinkaya-Rundel, and Garrett Grolemund is available at r4ds.hadley.nz and is especially helpful for learning the tidyverse.

- *Tidy Modeling with R* by Max Kuhn and Julia Silge is available at tmwr.org and introduces tidymodels.

For additional practice, try the weekly community data challenges from TidyTuesday.

The Internet

Consult slides, notes, and supplementary materials before searching online. If needed, try the Posit Community and then Stack Overflow. Employees are evaluated on the tools and coding style taught in job meetings, so online solutions may not meet job expectations.

Artificial Intelligence

Artificial intelligence (AI) is an important analytics tool, and employees may use it to understand and write code. Employees must also regulate their own use. Early-stage research suggests that active AI use may weaken learning during tasks such as writing or coding. We must balance the firm's productivity with each employee's intellectual and spiritual growth.

The gospel helps reconcile temporal and spiritual priorities. The following excerpt from "Things as They Really Are 2.0" offers useful perspective:

"Now, I believe AI appropriately can assist in gathering information, critiquing our thinking, evaluating our writing style, and accelerating an iterative process of learning "line upon line [and] precept upon precept." But the divine capacities to create and work belong uniquely to each of us as sons and daughters of God. We undoubtedly can generate and produce fabulous content for [school/church/work] with AI. But the objective is not merely producing or presenting impressive content; rather, it is working and becoming what God intends and yearns for us to become."

Keep these principles in mind:

- AI is a *tool*, neither good nor bad on its own. It can gather information and critique thinking, writing, or code, but it is not a source of truth and cannot replace personal revelation.
- Overreliance on AI can foster spiritual and temporal laziness. The work and learning process matters as much as the final product.
- AI cannot simulate or replace the influence of the Holy Ghost. Spiritual discernment remains essential.
- AI should supplement, not replace, human work, creativity, and divine capacities.

Remember - do not be "transformed from [employees] who can act into objects that are only acted upon."

Employee Success

Because employees must learn a programming language while learning to analyze data, **the beginning of employment has a steep learning curve**. Plan accordingly.

Use these practices to improve job performance:

1. Seek learning by study and faith (D&C 109:7).
2. Prepare for job meetings by previewing material and coming with questions.
3. Actively code, take notes, and ask questions.
4. Practice coding by completing exercises, referencing supplementary material as needed.
5. Review exercise solutions and note where and why your work differs.
6. Use knowledge checks to gauge your understanding.
7. Work with other employees and utilize office hours.
8. Download and organize all job materials, notes, and code.

Coding

Follow these coding practices:

1. Learn by doing: Code live during meetings and complete exercises.
2. Pay *careful* attention to details.
3. Don't code from scratch. Start with previous work and solutions.
4. Look at and emulate good code.
5. Literally sketch what transformed data should look like in the end.

Assessment

Performance Assessment (Grading) Philosophy

Performance assessment includes some subjectivity, so each deliverable uses a defined process:

- Skill-building exercises: employees evaluate their work with a provided rubric.
- Knowledge checks: employees complete written, in-person quizzes individually.
- Projects: each senior partner evaluates every open-ended project, and the average becomes the project's final evaluation. Employees who improve their project quality over the course of employment receive a performance boost.

Final performance assessments will follow a standard rubric. As mandated, the distribution of employee grades will result in no more than an A- average (*i.e.*, a 3.65 GPA).

A	93-100%	B-	80-82%	D+	67-69%
A-	90-92%	C+	77-79%	D	63-66%
B+	87-89%	C	73-76%	D-	60-62%

B	83-86%	C-	70-72%	E	0-59%

Final grades will be determined based on the following weights applied to the evaluations of each type of deliverable.

Exercises	20%
Knowledge checks	30%
Projects	50%

Employees may submit a **skill-building exercise** up to *one week* late for 75% credit. If an initial exercise score is below 75%, the employee may resubmit once for up to 75% credit. No exercise may be submitted more than twice. **Knowledge checks** and **projects** may not be submitted late without prior authorization from the CAO.

Skill-building Exercises

There are 13 skill-building exercises, due on the dates listed in the schedule. Exercise 3 combines the previously planned Exercises 3 and 4. Employees may collaborate but must submit their own work.

Employees will assess each exercise with the rubric in its posted solution and comment wherever their work differs from the solution. Add comments directly to R scripts or use the comment feature in Word documents.

Employees will have one week from the initial due date to submit their evaluated exercise. For exercises submitted during the week before Thanksgiving, graded versions are due two weeks after the initial due date, during the week after Thanksgiving. Students will receive a nominal amount of points for reviewing their exercises on time.

Knowledge Checks

Knowledge checks are written, in-person quizzes that assess comprehension of key analytics concepts and methods. Employees must complete them individually. Only printed or handwritten course materials are permitted; electronic devices and technological assistance are prohibited. There will be three knowledge checks, held on the dates listed in the schedule.

Projects

Each unit culminates in a project based on (hypothetical) client data and business problems. Employees have one week to produce a technical report and an executive summary of their results and recommendations. Employees may work in small groups but must submit individual work; identical or highly similar projects may be penalized. During project-week

meetings, employees can work together and consult the CAO and senior partners. Each employee will also review a colleague's project, documenting strengths and recommended improvements. Senior partners may recognize projects that exceed expectations with an extra commendation.

Extra Commendations

To support firm cohesion, employees may earn one additional point on their lowest project by completing a one-on-one, get-to-know-you meeting with the CAO. Office hours are subject to availability; use the scheduling link at the top of the syllabus.

Schedule

All deliverables except Project 3 are due on Canvas at the start of the associated meeting. In rows with two topics, the first is scheduled for Tuesday and the second for Thursday. Project 3 is due on Thursday, December 17, 2026, at 11:59 p.m. Mountain Time, the last day of finals.

The CAO may revise the syllabus, schedule, or company policies, including those governing AI use and performance assessment. Employees will receive reasonable notice of changes that affect deadlines.

Fall 2026 Semester Schedule

Week	Unit	Topic	Due Dates
Week 1 Sep. 3	Data	FAQ and the Analytics Process	
Week 2 Sep. 8, 10	Data	R and Transforming Data R and Transforming Data	Exercise 1 (T)
Week 3 Sep. 15, 17	Data	R and Transforming Data Visualize Discrete Data	
Week 4 Sep. 22, 24	Data	Visualize Continuous Data Tidy Data and Types	Exercise 2 (T) Exercise 3 (Th)
Week 5 Sep. 29, Oct. 1	Data	Codex Live Workshop	Exercise 4 (T) Knowledge Check 1 (T)
Week 6 Oct. 6, 8	Data	**Project 1: iFood**	
Week 7 Oct. 13, 15	Inference	Linear Models Evaluate Model Fit	Project 1 (T) Exercise 5 (Th)
Week 8 Oct. 20, 22	Inference	Explanatory Variables Overfitting and Prediction	Exercise 6 (T) Exercise 7 (Th)
Week 9 Oct. 27, 29	Inference	Preprocess Data Preprocess Data	Exercise 8 (T)
Week 10 Nov. 3, 5	Inference	Integrity and Causal Inference Live Workshop	Exercise 9 (T)
Week 11 Nov. 10, 12	Inference	**Project 2: Mile High Stays**	Knowledge Check 2 (T)
Week 12 Nov. 17, 19	Prediction	Logistic Regression Decision Trees	P2: Executive Summary (T) Exercise 10 (Th)
Week 13 Nov. 24, 26	–	**No class: Thanksgiving week**	
Week 14 Dec. 1, 3	Prediction	Random Forests K-Means Clustering	Exercise 11 (T) Exercise 12 (Th)
Week 15 Dec. 8, 10	Prediction	Live Workshop **Project 3: Client TBD**	Exercise 13 (T) Knowledge Check 3 (Th)

Project 3 due: Thursday, December 17, 2026, at 11:59 p.m. Mountain Time.

Honor Code

In keeping with the principles of the BYU Honor Code, employees are expected to be honest in all of their work. Academic honesty means, most fundamentally, that any work you present as your own must in fact be your own work and not that of another. Violations of this principle may result in a failing grade on the deliverable and/or in the course and additional disciplinary action by the university. Employees are also expected to adhere to the Dress and Grooming Standards. Adherence demonstrates respect for yourself and others and ensures an effective learning and working environment. It is the university's expectation, and every instructor's expectation, that each student will abide by all Honor Code standards. Please call the Honor Code Office at 801-422-2847 if you have questions about those standards.

Preventing & Responding to Sexual Misconduct

Brigham Young University prohibits all forms of sexual harassment – including sexual assault, dating violence, domestic violence, and stalking on the basis of sex – by its personnel and students and in all its education programs or activities. University policy requires all faculty members to promptly report incidents of sexual harassment that come to their attention in any way and encourages reports by students who experience or become aware of sexual harassment. Incidents should be reported to the Title IX Coordinator at title9@byu.edu or 801-422-8692 or 1085 WSC. Reports may also be submitted online at titleix.byu.edu/report or 1-888-238-1062 (24-hours a day). BYU offers a number of resources and services for those affected by sexual harassment, including the university's confidential Sexual Assault Survivor Advocate. Additional information about sexual harassment, the university's Sexual Harassment Policy, reporting requirements, and resources can be found in the University Catalog, by visiting titleix.byu.edu, or by contacting the university's Title IX Coordinator.

Student Disability

Brigham Young University is committed to providing a working and learning atmosphere that reasonably accommodates qualified persons with disabilities. A disability is a physical or mental impairment that substantially limits one or more major life activities. Whether an impairment is substantially limiting depends on its nature and severity, its duration or expected duration, and its permanent or expected permanent or long-term impact. Examples include vision or hearing impairments, physical disabilities, chronic illnesses, emotional disorders (*e.g.*, depression, anxiety), learning disorders, and attention disorders (*e.g.*, ADHD). If you have a disability which impairs your ability to complete this course successfully, please contact the University Accessibility Center (UAC), 2170 WSC or 801-422-2767 to request a reasonable accommodation. The UAC can also assess students for learning, attention, and emotional concerns. If you feel you have been unlawfully discriminated against on the basis of disability, please contact Shawn Smith at the Equal Employment Office at 801-422-2845, D-276 ASB for help.

Inappropriate Use of Course Materials

All course materials (*e.g.*, outlines, handouts, syllabi, exams, quizzes, PowerPoint presentations, lectures, audio and video recordings, etc.) are proprietary. Students are prohibited from posting or selling any such course materials without the express written permission of the professor teaching this course. To do so is a violation of the Brigham Young University Honor Code.

Academic Honesty

One of the injunctions of the Honor Code is the call to "be honest." Students come to the university not only to improve their minds, gain knowledge, and develop skills that will assist them in their life's work, but also to build character. "President David O. McKay taught that character is the highest aim of education. . . " (The Aims of a BYU Education). It is the purpose of the BYU Academic Honesty Policy to assist in fulfilling that aim. BYU students should seek to be totally honest in their dealings with others. They should complete their own work and be evaluated based upon that work. They should avoid academic dishonesty and misconduct in all its forms, including but not limited to plagiarism, fabrication or falsification, cheating, and other academic misconduct.

Marriott School of Business Inclusion Statement

BYU Marriott is committed to fostering an environment of belonging for all students and employees. In accordance with the BYU Marriott vision, mission, and values, "We value respect for all individuals as children of God and recognize the inherent worth, divine potential, and agency of each person. A climate of respect and belonging enhances our learning, facilitates collaboration, and encourages personal growth."

We embrace the statement President Russell M. Nelson made on June 1, 2020.

"The Creator of us all calls on each of us to abandon attitudes of prejudice against any group of God's children. Any of us who has prejudice toward another race needs to repent!

During the Savior's earthly mission, He constantly ministered to those who were excluded, marginalized, judged, overlooked, abused, and discounted. As His followers, can we do anything less?

Let us be clear. We are brothers and sisters, each of us the child of a loving Father in Heaven. His Son, the Lord Jesus Christ, invites all to come unto Him – 'black and white, bond and free, male and female,' (2 Nephi 26:33). It behooves each of us to do whatever we can in our spheres of influence to preserve the dignity and respect every son and daughter of God deserves."

Mental Health

Mental health concerns and stressful life events can affect students' academic performance and quality of life. BYU Counseling and Psychological Services (CAPS, 1500 WSC, 801-422-3035, caps.byu.edu) provides individual, couples, and group counseling, as well as stress management services. These services are confidential and are provided by the university at no cost for full-time students. For general information please visit caps.byu.edu; for more immediate concerns please visit help.byu.edu.